Venue: ..... RollNo.: .....
Copy : ..... Name: .....

INSTRUCTIONS:

(i) As mentioned in class, the word "differentiable" is a synonym for "smooth": In single variable case, derivatives of all orders exist and are continuous while in multi–variable case, partial derivatives of all orders exist and are continuous.

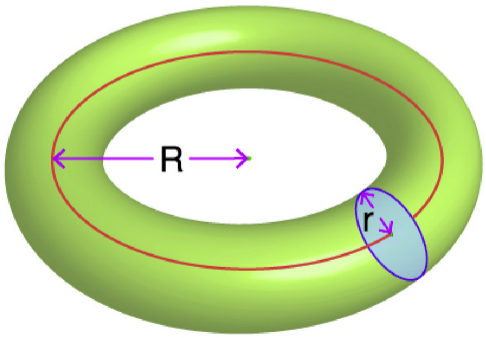
(ii) Show all steps. Calculators are not permitted.

(iii) There are ten questions on this test, each worth 4 marks. Additionally, there is an optional jackpot question.

(iv) Answer all questions in designated space or over the extra pages towards the end of the booklet. Use supplementary sheets ONLY for rough work. DO NOT TIE supplementary sheets to this answer booklet.

Q-11) JACKPOT: {Full solution begets an A on this test. No partial marks. Answer on the last page.}

Consider two torii in $\mathbb{R}^3$, of the form shown. One has $R = 14, r = 2$ while the second one has $R = 7, r = 4$. Assume that the measurements are in centimetres. Are the two isometric?



QUESTIONS:

Q-1) For real numbers $r > 0$ and p, a helix $\alpha : \mathbb{R} \to \mathbb{R}^3$ is given as $\alpha(t) = (r\cos t, r\sin t, pt)$. Find the curvature and torsion of α.

Venue: ..... RollNo.: .....
Copy : ..... Name: .....

Q-2) For a positive real number L, consider an arc–length parametrized smooth curve $\alpha : [0, L] \to \mathbb{R}^3$. You are given that α is without singular points of orders 0 and 1. If the the torsion of α is identically zero, show that the trace of α is contained in a plane.

Venue: ..... RollNo.: .....
Copy : ..... Name: .....

Q-3) For a positive real number L, let $\kappa : [0, L] \to \mathbb{R}$ be a given differentiable function. Find an arc–length parametrized plane curve $\alpha : [0, L] \to \mathbb{R}^2$ whose curvature function is κ.

Q-4) For a 3–tuple $t = (a, b, c) \in \mathbb{R}^3$ we define the solution set

$$S_t = \{(x, y, z) | ax^2 + by^2 + cz^2 = 1\}.$$

Find all t for which S_t is a regular surface.

Venue: ..... RollNo.: .....

Copy : ..... Name: .....

Q-5) Find all isometries of the 3–dimensional euclidean space $\mathbb{R}^3$.

Venue: ..... RollNo.: .....

Copy : ..... Name: .....

Q-6) Let $x : U \to S$ and $y : V \to S$ be two systems of co–ordinates for a regular surface S. Let $p \in x(U) \cap y(V)$. Find the relation between coefficients of the first fundamental form at p as expressed in these two systems of co–ordinates.

Venue: ..... RollNo.: .....

Copy : ..... Name: .....

Q-7) On a bounded open set $Q \subset \mathbb{R}^2$, let $f : Q \to \mathbb{R}$ be a given smooth function. Define $R = \{(x, y, z) | (x, y) \in Q,\ z = f(x, y)\} \subset \mathbb{R}^3$. Show that the area of R is given by

$$\int\int_Q \sqrt{1 + f_x^2 + f_y^2}\ dxdy.$$

Q-8) For two real numbers $r > a > 0$, consider a circle of radius r in the y–z plane centred at $(0, a, 0)$. It is rotated about the z axis to obtain a torus. An ant starts walking from a point on the torus such that its latitude and longitude change with time uniformly, and, both at the same rate. Set up the integral to find the distance covered when the ant returns to the starting point.

Venue: ..... RollNo.: .....
Copy : ..... Name: .....

Q-9) From a right circular cone of base radius r and height h, delete the base circle and the apex† point to obtain a clown's hat H. Briefly check that the clown's hat is a regular surface. Find the area of H using the methods taught in our course.
{†: *top–most* }

Venue: ..... RollNo.: .....

Copy : ..... Name: .....

Q-10) For a regular surface S, let $x : U \to S$ be a system of co–ordinates around a point $p \in S$. In this system of co–ordinates, suppose you calculate and find the gaussian curvature of S at p to be the real number K. What is the gaussian curvature at the same point wrt the system of co–ordinates obtained by exchanging the input variables to x?
{Let $\tilde{U} = \{(s,t)|(t,s) \in U\}$ and $\tilde{x} : \tilde{U} \to S$ be given by $\tilde{x}(s,t) = x(t,s)$. Use $\tilde{N} = \frac{\tilde{x}_s \times \tilde{x}_t}{\|\tilde{x}_s \times \tilde{x}_t\|}$ as the new normal unit vector field.}

Venue: ..... RollNo.: .....
Copy : ..... Name: .....

Q-?)

Xtra space for answers, NOT rough work.

Q-?)

Xtra space for answers, NOT rough work.

Venue: ..... RollNo.: .....
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Q-?)

Xtra space for answers, NOT rough work.

Q-?)

Xtra space for answers, NOT rough work.